CAMARINES SUR POLYTECHNIC COLLEGES
COLLEGE of COMPUTER STUDIES

CSAC 212 - Introduction to Artificial Intelligence

Early Stage Diabetes Risk Prediction: A Predictive Modeling of Occurrence of Diabetes Using XGBoost

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Abstract

Diabetes is a life-threatening chronic disease that is imminent throughout the world. International Diabetes Federation (IDF) reports that about 537 million people were diagnosed worldwide, the Philippines being one of them. An early diagnosis and management are essential for delaying progression. Thus, the researcher proposes a project to develop a model for efficiently predicting the risk of early-stage diabetes. The study uses a decision tree-based algorithm, Extreme Gradient Boosting (XGBoost), to predict whether a patient is at risk. The dataset's inputs contain the signs and symptoms of newly diagnosed or diabetic patients.

1 Introduction

Diabetes is among the top killer diseases in the Philippines. It is a chronic disease in which the pancreas either lacks or fails to produce enough insulin, or for one reason or another, the body cannot make effective use of that insulin produced. International Diabetes Federation (IDF) reports that 537 million individuals were diagnosed with diabetes worldwide and 206 million in the Western Pacific Region. It is estimated that the number will rise to 260 million by 2045. In the Philippines, there are about a total of 66 million adult population; 4.3 million of this population were diagnosed with diabetes, and 7.5% of the adult population is prevalence [13]. Knowing the possibility of having an early stage of diabetes is essential for early treatment and management.

The input of the program consists of a labeled data set of demographic information (age and sex) and diabetes-related symptoms (such as polyuria, polydipsia, sudden weight loss, weakness, polyphagia, genital thrush, visual blurring, itching, irritability, delayed healing, partial paresis, muscle stillness, alopecia and obesity), where most features are binary (Yes/No), except for the age which is continuous (20-65) and sex which is categorical (Male/Female). The dataset can be used to train eXtreme Gradient Boosting (XGBoost) for early-stage diabetes risk prediction. The expected output is a prediction model for early-stage diabetes, trained using the decision tree-based algorithm XGBoost.

2 Related work

The proponent gathered the following research studies from Google Scholar, Academia, IEEE Xplore, ScienceDirect, IOP, Springer, and Research Gate to assist in the present study. These studies are applicative machine learning algorithms that have been applied to many publications and are connected to the dataset.

According to Sheridan et al. [16], the proposed algorithm is better than Random Forest, and it is as good as neural networks since XGBoost has many adjustable parameters that can define a set of standard parameters on which the model makes predictions. Speed is said to be the biggest strength of XGBoost.

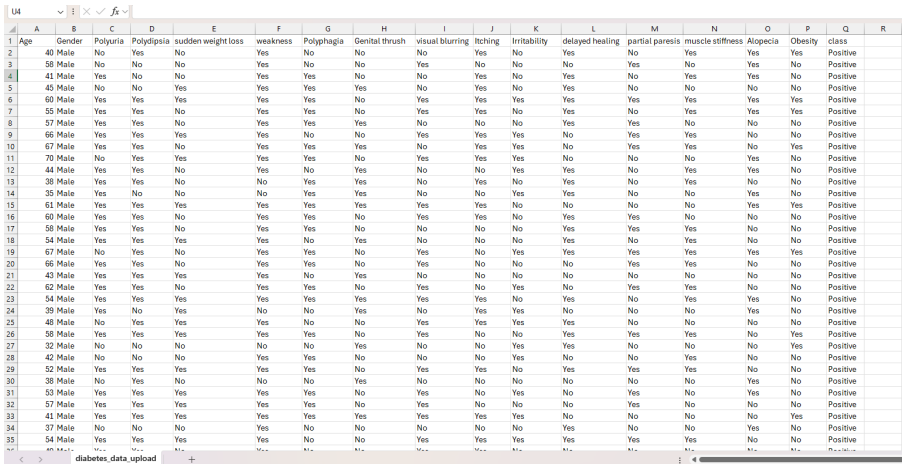
In a recent study by Stokanović et al. [17], on the classification task of breast cancer and heart disease, XGBoost was robust against missing data that resulted in less than 2.5% in degradation using simple imputation methods. Results, therefore, suggest a validation of research for the effectiveness of XGBoost in medical prediction tasks. Furthermore, the study validated using XGBoost in early-stage diabetes prediction since the algorithm has proven its strength in handling incomplete medical data and its efficiency in disease classification tasks. Good F1-scores are preserved even with high missing instances up to 90% (94.1% for breast cancer and 73% for heart failure data), and stability across different algorithm learning parameters is presented.

Guan et al. [7] conducted a study that predicts the early occurrence of lung cancer through an interdisciplinary approach combining metabolomics and XGboost. They selected carnitine indicators and demographic indicators of the subject, and surprisingly, their study garnered 0.81 AUC, 75.29 accuracy, and 74 sensitivity, showing the superior predictive power of XGBoost.

In the study of cervical cancer risk prediction by Jha et al. [10], XGBoost showed stronger potentialities. According to their study, XGBoost's sparse-aware and quantile algorithms utilized inbuilt algorithms for handling missing data with more potency than traditional machine learning models and made effective parallel computability. Their study proved fruitful in validating a strength of XGBoost—the ability to find relevant predictors for diseases to predict early stages that exactly mirrors the needs of early-stage prediction of diabetes. A similar study by Ryu et al. [14] on using XGBoost in dementia risk prediction highlighted its effectiveness in processing medical data and health conditions. An optimized model they developed achieved high results (with 85.61% accuracy and 79.28% F1-score) in contrast to other ML algorithms. The extraction of important variables and generating derived variables, combined with hyperparameter optimization, have given valuable insights for the current study.

Khafaga et al. [11] conducted a similar study. Their approach incorporates detection and balancing data distribution techniques. They then utilized four classification algorithms to develop a prediction model based on actual data. Results revealed that K Nearest Neighbor (KNN) yielded the highest accuracy of 97.36%. Another study by Sadhu and Jadli [15] took a similar approach. Their study implemented seven machine learning classifiers: KNN, LR, DT, RF, SVM, NB, and ANN, on the same dataset used in the current study and three different evaluation metrics: classification accuracy, F-score, and ROC. Surprisingly, Random Forest favors an average-sized dataset as their study results. The study of Islam et al. [9] revealed that random forest is the best algorithm for the dataset and has the best accuracy in their study. They analyzed the dataset together with Logistic Regression and Naive Bayes and applied validation and evaluation techniques. Another similar study by Kumar et al. [12] utilized Gradient Boosting Decision Tree (GBDT); CatBoost, an ensemble approach for predicting diabetes risk at early stages. They also compared it to other machine learning methods, resulting in the model outperforming them. In 2021, V N Wijayaningrum et al. [18] conducted a similar study in predicting the risk of early-stage diabetes using Multi-Layer Perceptron (MLP), with optimal parameters obtained from optimization results of the proposed algorithm. The study yielded an accuracy of 97.69% and 96.92% for one and two hidden layers.

Many articles and studies were found, but only a few were gathered for this study as they can give ideas and guidance to the researcher. The comparisons and tests conducted in previous publications help the proponent of this study to go further into what was already known and improve the methods of predicting the risk of early-stage diabetes.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Age	Gender	Polyuria	Polydipsia	sudden weight loss	weakness	Polyphagia	Genital thrush	visual blurring	Itching	Irritability	delayed healing	partial paresis	muscle stiffness	Alopecia	Obesity	class	
2	40	Male	No	Yes	No	Yes	No	No	No	Yes	No	Yes	No	Yes	Yes	Yes	Positive	
3	58	Male	No	No	No	Yes	No	No	Yes	No	No	No	Yes	No	Yes	No	Positive	
4	41	Male	Yes	No	No	Yes	Yes	No	No	Yes	No	Yes	No	Yes	Yes	No	Positive	
5	45	Male	No	No	Yes	Yes	Yes	Yes	No	Yes	No	Yes	No	No	No	No	Positive	
6	60	Male	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Positive	
7	55	Male	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Positive	
8	57	Male	Yes	Yes	No	Yes	Yes	No	No	No	No	Yes	Yes	No	No	No	Positive	
9	66	Male	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	Yes	Yes	No	No	Positive	
10	67	Male	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Positive	
11	70	Male	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	Yes	No	Positive	
12	44	Male	Yes	Yes	No	Yes	No	Yes	No	No	Yes	Yes	No	Yes	Yes	No	Positive	
13	38	Male	Yes	Yes	No	No	Yes	Yes	No	Yes	No	Yes	No	Yes	No	No	Positive	
14	35	Male	Yes	No	No	No	Yes	Yes	No	No	Yes	Yes	No	No	Yes	No	Positive	
15	61	Male	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Yes	Positive	
16	60	Male	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	No	No	Positive	
17	58	Male	Yes	Yes	No	Yes	Yes	No	No	No	No	Yes	Yes	Yes	No	No	Positive	
18	54	Male	Yes	Yes	Yes	Yes	No	Yes	No	No	No	Yes	No	Yes	No	No	Positive	
19	67	Male	No	Yes	No	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Positive	
20	66	Male	Yes	Yes	No	Yes	Yes	No	Yes	No	No	No	Yes	Yes	No	No	Positive	
21	43	Male	Yes	Yes	Yes	Yes	No	Yes	No	No	No	No	No	No	No	No	Positive	
22	62	Male	Yes	Yes	No	Yes	Yes	No	Yes	No	Yes	No	Yes	Yes	No	No	Positive	
23	54	Male	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	Yes	Yes	No	Positive	
24	39	Male	Yes	No	Yes	No	No	Yes	No	Yes	Yes	No	No	No	Yes	No	Positive	
25	48	Male	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	No	No	No	No	Positive	
26	58	Male	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	Yes	Yes	Yes	No	Yes	Positive	
27	32	Male	No	No	No	No	No	Yes	No	No	Yes	Yes	No	No	No	Yes	Positive	
28	42	Male	No	No	No	Yes	Yes	No	No	No	Yes	No	No	Yes	No	No	Positive	
29	52	Male	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	No	No	Positive	
30	38	Male	No	Yes	No	No	No	Yes	No	No	No	No	No	No	Yes	No	Positive	
31	53	Male	Yes	Yes	Yes	Yes	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Positive	
32	57	Male	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No	Yes	No	No	No	Positive	
33	41	Male	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	Yes	Positive	
34	37	Male	No	No	No	Yes	No	No	No	No	No	Yes	No	No	Yes	No	Positive	
35	54	Male	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Positive	

Figure 1: The dataset containing diabetes-related symptoms

3 Methods

eXtreme Gradient Boosting (XGBoost) implements the gradient boosting framework quite efficiently. [6][5]. The package comes with efficient linear model solvers and tree learning algorithms and supports other objective functions, including classification [1] like the current study. This is one of the most accurate implementations of gradient boosting, primarily designed to boost the efficiency and efficacy of machine learning models, up to the limit of computation capacity by boosted tree methods. Unlike the Gradient Boosting Decision Trees (GBDT), XGBoost constructs trees simultaneously. Instead of calculating split quality at each possible split over the training set, it takes a level-wise approach by scanning all gradient values and makes use of such partial sums[19]. Artificial neural networks are traditionally much better at prediction problems, especially in unstructured data, such as pictures and text. Nowadays, for structured/tabular-sized small to medium-sized datasets, the best performing is based on decision trees.[8].

3.1 The Dataset

This study uses the dataset from UCI, ML Repository entitled "Early Stage Diabetes Risk Prediction," gathered by a survey from the patients of Sylhet Diabetes Hospital in Bangladesh[4]. The dataset is also linked to the "Early Stage Diabetes Risk Prediction Dataset" from Kaggle Repositories[3]. The dataset consists of 16 inputs and one output. It is indicated that the data has missing values. Accordingly, the algorithm takes inputs consisting of labeled data of the patient's age, sex, and demographic information. The output class consists of 'positive' and 'negative' values. We break our data into three main sets: the set for training and fitting the model, the set for validating the model so it does not fit too well with the data used for training, and the set for testing the validation of the performance of the trained models. 80% will be allocated for training and 20% for testing data. Moreover, the testing data will divided equally to allocate a portion for validation and testing. The dataset can then be used to train a decision tree-based algorithm, XGBoost, in predicting the risk of early-stage diabetes.

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